Lower intake of vegetables and legumes associated with cognitive decline among illiterate elderly Chinese: a 3-year cohort study.

BACKGROUND: Despite many studies on cognitive function and its influential factors among old population, relatively little research has been designed to study the relationship between dietary intake and cognitive function in elderly. OBJECTIVE: We conducted a population-based, prospective nested case-control study to investigate the association between dietary habits and declines in cognitive function over three years among Chinese illiterate elderly. DESIGN AND METHODS: This study was part of the Chinese Longitudinal Health Longevity Study (CLHLS). Six thousand nine hundred and eleven illiterate residents aged 65 or older were investigated. Socio-demographic and dietary habits data were collected at baseline. The cognitive function of illiterate elderly persons was assessed using Chinese revised Mini Mental State Examination (MMSE-r) in 2002 and 2005. Cognitive decline was defined as MMSE-r score dropped to less than 18 at follow-up among those with normal cognitive function (MMSE-r≥18 at baseline). Odds ratios (OR) were calculated via logistic regression models. RESULTS: Five thousand six hundred and ninety one elderly were included in the current analysis. In bivariate analysis, cognitive decline was associated with gender, marital status ,financial status, smoking, drinking alcohol, drinking tea, eating fruits, vegetables, legumes, fishes, meat, egg and sugar. Multivariate logistic regression analysis found that always eating vegetable (Adjusted OR: 0.66; 95% confidence intervals, CI: 0.58, 0.75), always consuming legumes (AOR:0.78; 95% CI: 0.64, 0.96) were inversely associated with cognitive decline. CONCLUSIONS: Lower intakes of vegetables and legumes were associated with cognitive decline among illiterate elderly Chinese. Dietary factors may be important for prevention cognitive decline.